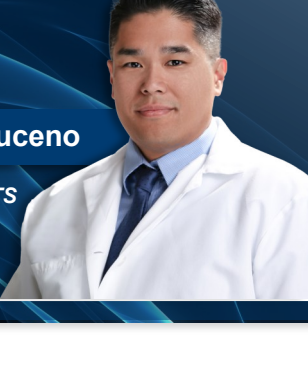




Low Back Pain Imaging Guidelines for Virtual Visits

Core Imaging Principles

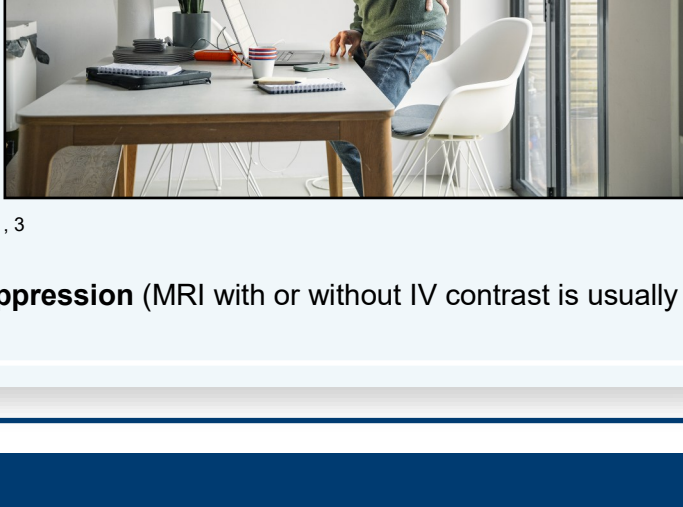
Imaging is usually not appropriate for patients with acute or chronic low back pain without red flags, regardless of visit modality.¹ The same imaging indications and non-indications apply to virtual visits as to in-person encounters. Most cases of uncomplicated low back pain are self-limited and responsive to medical management and physical therapy.²

Imaging Indications

When to Order Imaging

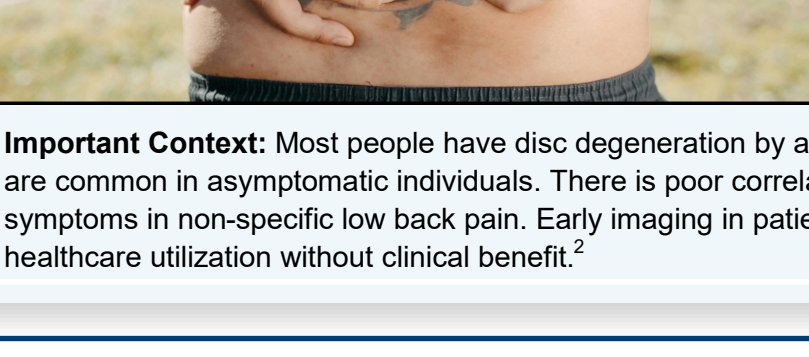
In the virtual setting, if imaging is being considered, an in-person evaluation is likely warranted to determine appropriate further evaluation.

- Persistent symptoms after 6 weeks of optimal medical management** in patients who are candidates for surgery or intervention (MRI lumbar spine without IV contrast is usually appropriate)¹
- Suspected cauda equina syndrome** (MRI with or without IV contrast is usually appropriate)^{1, 3}
- History of prior lumbar surgery** with new or progressing symptoms (radiography or MRI is usually appropriate)¹
- Low-velocity trauma, osteoporosis, elderly age (≥75 years), or chronic steroid use** (radiography, MRI, or CT is usually appropriate)^{1, 3}
- Suspicion of cancer, infection, or immunosuppression** (MRI with or without IV contrast is usually appropriate)^{1, 3}



Imaging Non-Indications

When NOT to Order Imaging



- Nonspecific low back pain without red flags (acute, subacute, or chronic)^{1, 2}
- Early presentation (6 weeks) without red flag findings^{1, 2}
- Patients without neurologic compromise who have minor risk factors (consider imaging only after trial of therapy)¹

Important Context: Most people have disc degeneration by age 40 years, and imaging abnormalities are common in asymptomatic individuals. There is poor correlation between imaging findings and symptoms in non-specific low back pain. Early imaging in patients without red flags leads to increased healthcare utilization without clinical benefit.²

Red Flag Assessment for Virtual Visits

Most red flags can be effectively assessed through history alone during telemedicine encounters and guided to the clinically-appropriate in-person evaluation.⁴

Red Flags Identifiable by History	
Red Flag	Symptoms
Cauda Equina Syndrome	- New-onset urinary retention or fecal incontinence - Saddle anesthesia (patient-reported) - Severe or progressive lower extremity neurologic deficit
Malignancy	- History of cancer with new-onset low back pain - Unexplained weight loss - Failure to improve after 1 month - New onset at age >50 years
Infection	- Fever - Immunosuppression - IV drug use - Recent infection, indwelling catheters, central lines
Fracture	- Significant trauma - Minor trauma with osteoporosis, chronic corticosteroid use, or age ≥75 years

Virtual Physical Examination Techniques

Pre-Visit Patient Preparation

Instruct patients to:

- Wear shorts or loose-sitting clothing
- Position camera at waist height, 6-8 feet away
- Have available: sturdy chair, wall for support, household items (e.g., water bottle, book)
- Ensure adequate lighting and clear floor space
- Have family member present if balance is a concern

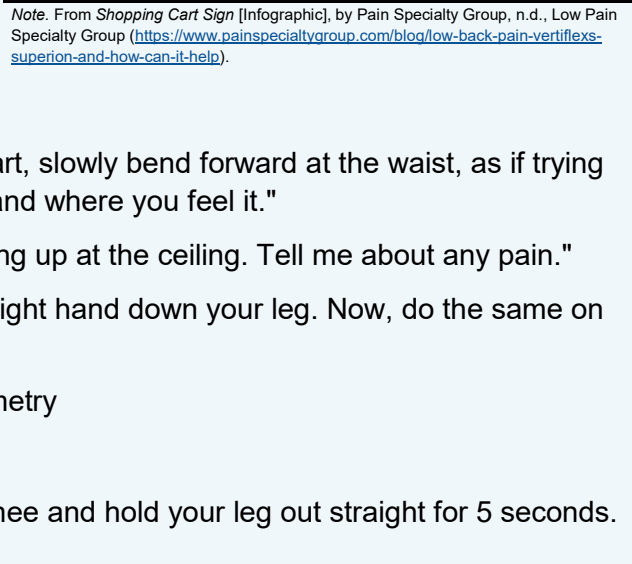
Observation and Gait Assessment ^{3, 5}

Script: "Stand facing the camera with your arms at your sides. Now, turn to show me your side profile, then your back."

Observe for: posture, spinal alignment, muscle asymmetry, flexed positioning (shopping cart sign, suggesting spinal stenosis)

Script: "Walk away from the camera, turn around, and walk back toward me."

Observe for: gait abnormalities, antalgic gait, foot drop



Note: From Shopping Cart Sign [Infographic], by Pain Specialty Group, n.d., Low Pain Specialty Group (<https://www.painspecialtygroup.com/blog/low-back-pain-vertiflex-superior-and-how-can-it-help>).

Range of Motion Testing

Flexion: "Standing with your feet a shoulder's-width apart, slowly bend forward at the waist, as if trying to touch your toes. Tell me if this reproduces your pain and where you feel it."

Extension: "Stand up straight and lean backward, looking up at the ceiling. Tell me about any pain."

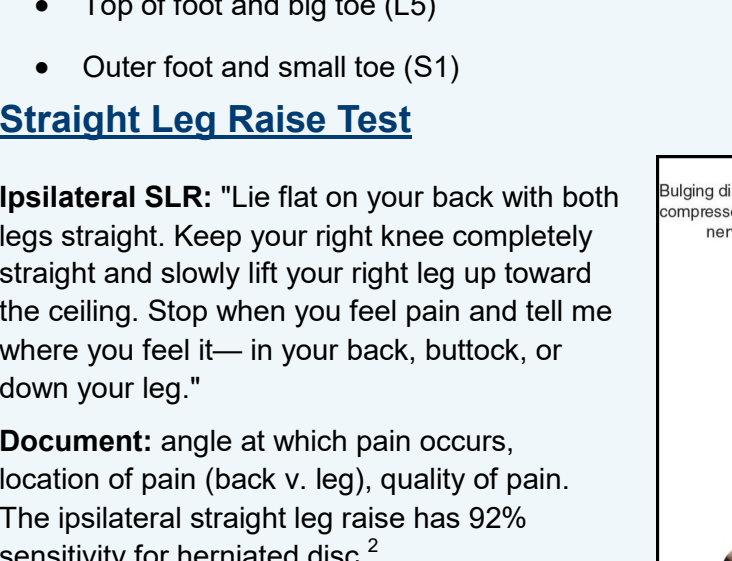
Lateral Bending: "Bend to your right side, sliding your right hand down your leg. Now, do the same on the left side."

Document: degree of motion, pain reproduction, asymmetry

Neurologic Examination: Motor Strength

L4 (Quadriceps): "Sit in a chair. Straighten your right knee and hold your leg out straight for 5 seconds. Now, do the same with your left leg."

Alternative: "Stand on one leg for 5 seconds. This tests the strength of your thigh muscles."



(1A) demonstration of regular ambulation; (1B) walking on toes, gastrocnemius and soleus (S1, S2); (1C) walking on heels, tibialis anterior (L4, L5); (1D) tandem gait to assess balance.

Note: From Lower Extremity Motor Examination [Photograph], by Sardar, Z. M., Coury, J. R., et al., 2021, World Neurosurgery (<https://pubmed.ncbi.nlm.nih.gov/34237452/>).

resistance. Push down on your own knee with your hands to create resistance. Now, do the same with your left leg."

Sensory Testing

Script: "Using your finger, trace the area where you feel numbness or tingling. On camera, show me where this is located."

Dermatomal Assessment: "I'm going to describe different areas of your leg. Tell me if each area feels normal, numb, or tingly compared to the other leg:"

- Front of thigh (L2-L3)
- Inner shin and knee (L4)
- Top of foot and big toe (L5)
- Outer foot and small toe (S1)

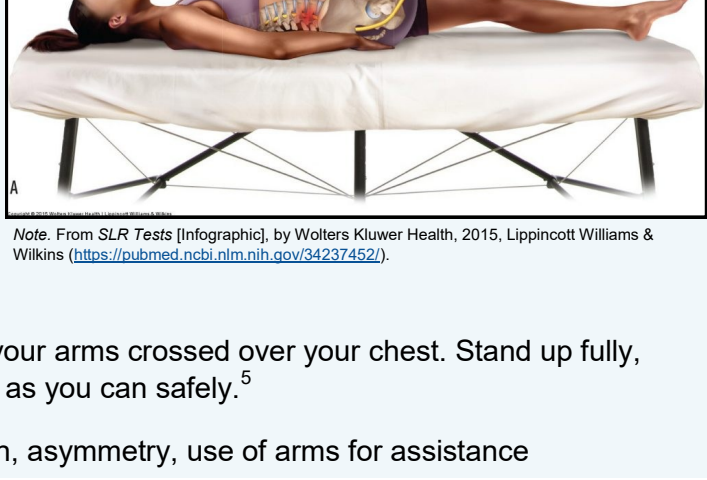
Straight Leg Raise Test

Ipsilateral SLR: "Lie flat on your back with both legs straight. Keep your right knee completely straight and slowly lift your right leg up toward the ceiling. Stop when you feel pain and tell me where you feel it— in your back, buttock, or down your leg."

Document: angle at which pain occurs, location of pain (back v. leg), quality of pain. The ipsilateral straight leg raise has 92% sensitivity for herniated disc.²

Contralateral SLR: "Now, keep your right leg flat and lift your left leg. Tell me if this causes pain your right leg or back."

Note: contralateral SLR is 90% specific for herniated disc.²



Note: From SLR Tests [Infographic], by Wolters Kluwer Health, 2015, Lippincott Williams & Wilkins (<https://eubmed.ncbi.nlm.nih.gov/34237452/>).

Functional Testing

Five Repetitive Sit-to-Stand: "Sit in a chair with your arms crossed over your chest. Stand up fully, then sit back down. Repeat this 5 times as quickly as you can safely."⁵

Observe for: ability to complete, pain reproduction, asymmetry, use of arms for assistance

Limitations of Virtual Assessment

Components that Cannot be Reliably Assessed Virtually ⁴		
Deep tendon reflexes (patellar, Achilles)	Rectal tone and perianal sensation	Palpation for tenderness, spasm, or step-off deformities
Precise sensory dermatomal mapping	Muscle bulk/atrophy assessment	Subtle motor weakness (may miss 4/5 strength)

When to Transition to In-Person Evaluation

Recommend an in-person assessment when:⁴

- Red flags identified that require physical confirmation (e.g., suspected cauda equina with need for rectal tone assessment, palpable abdominal mass)
- Diagnosis or management plan remains uncertain after virtual assessment
- Progressive or severe neurologic deficits reported
- Possible deformity or neurovascular compromise
- Symptoms worsen or fail to improve with initial management

Physical Therapy

Encourage patients to utilize physical therapy. To order physical therapy, use the **.APT SmartPhrase**.

Name	Description
☆ APTAV	Access to PT (APT) Antelope Valley
☆ APTAVSPANISH	Access to PT (APT) Antelope Valley Spanish
☆ APTBP	Access to PT (APT) Baldwin Park
☆ APTBPSPANISH	Access to PT (APT) Baldwin Park Spanish
☆ APTDO	Access to PT (APT) Downey
☆ APTDOSPANISH	Access to PT (APT) Downey Spanish
☆ APTDOUSLCERNOTEFP	Patient presents with pain in mouth for *** { :10918}. Associa...
☆ APTKC	Access to PT (APT) Kern County
☆ APTKCSPANISH	Access to PT (APT) Kern County Spanish

I'm ordering physical therapy to address your [***]. Please call (323) 783-2972 to schedule a Physical Therapy appointment.

APT Exercises -

☐	ACHILLES TENDINITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	ACUTE KNEE PAIN APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	ACUTE LOW BACK PAIN APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	ACUTE NECK PAIN APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	ANKLE SPRAIN APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	CARPAL TUNNEL SYNDROME APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	CRIOGENIC HEADACHE APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	FROZEN SHOULDER (STIFF) APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	GOLFER'S ELBOW / MEDIAL EPICONDYLITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	HIP ARTHRITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	HIP BURSITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	HIP STRAIN / SPRAIN APT (SmartText: 210856) PHYSICAL THERAPY) PT AMB SCAL
☐	KNEE ARTHRITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	PATELLO-FEMORAL SYNDROME APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	PLANTAR FASCIITIS APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	SHOULDER IMPINGEMENT APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL
☐	TENNIS ELBOW APT (ACCESS TO PHYSICAL THERAPY) PT AMB SCAL

I'm ordering physical therapy to address your [***]. Please call (323) 783-2972 to schedule a Physical Therapy appointment.

APT Exercises: Bridge use link below or scan QR code for Acute Low Back Pain exercises

https://www.medbridgego.com/access_token/HNC4FHX3

English Language

You also have the option of utilizing the spine E-Visit via KP.ORG app or website for additional care.

This QR code will take you to the KP app on your phone or go to kp.org and select the Back and Neck Pain E-visit.

NOEL SAN JOSE NEPOMUCENO MD

Documentation Template for Virtual Visits

Document:

- Specific red-flag questions asked and responses
- Patient-performed examination maneuvers and findings
- Limitations of virtual examination
- Clinical decision-making regarding imaging
- Plan for in-person follow-up if needed

Evidence Supporting Virtual Assessment

Telehealth spine examination components including inspection, lower extremity muscle testing, pain localization, straight leg raise, and sit-to-stand testing, demonstrating good reliability and validity compared to in-person examination.⁵

Appendix

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